Bayesian Network Inference for Resolving Uncertainty in Financial Depth–Stability Links

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Abstract. The relationship between financial depth and banking stability remains contested due to unresolved uncertainty about confounding factors and transmission mechanisms. This paper addresses two questions: whether GDP per capita confounds the depth–stability link, and whether effects operate through indirect lending pathways. We apply Bayesian network inference—combining the PC algorithm, expert-guided network construction, and bootstrap validation—to $N = 285$ country-year observations from ten ASEAN economies (2005–2020). Three findings emerge. First, GDP per capita is a dominant confounder ($\mathrm{BF}_{10} = 818{,}156$); after conditioning, financial depth shows a negative association with stability ($\hat{\beta} = -0.130$, $t = -6.79$). Second, the PC algorithm detects no direct financial depth–stability edge. Third, the hypothesized indirect pathway (FD $\rightarrow$ NIM $\rightarrow$ FDI $\rightarrow$ NPL $\rightarrow Z$) is structurally present but transmits a negligible effect ($\hat{\theta} = 6.12\times10^{-4}$, 95% CI contains zero). GDP confounding emerges as the primary source of inconsistency, demonstrating that structural pathway stability does not guarantee quantitative significance.

Keywords: Bayesian network · uncertainty resolution · financial depth · bank stability · GDP confounding · ASEAN

1 Introduction

Financial depth, measured as private credit to GDP, has a non-monotonic relationship with economic outcomes: credit deepening boosts growth until a threshold of 80–100% of GDP, beyond which further expansion may destabilize the banking system [3,20]. Deposit insurance–induced moral hazard is one channel, as protected depositors weaken market discipline [12,2]. Nearly all studies

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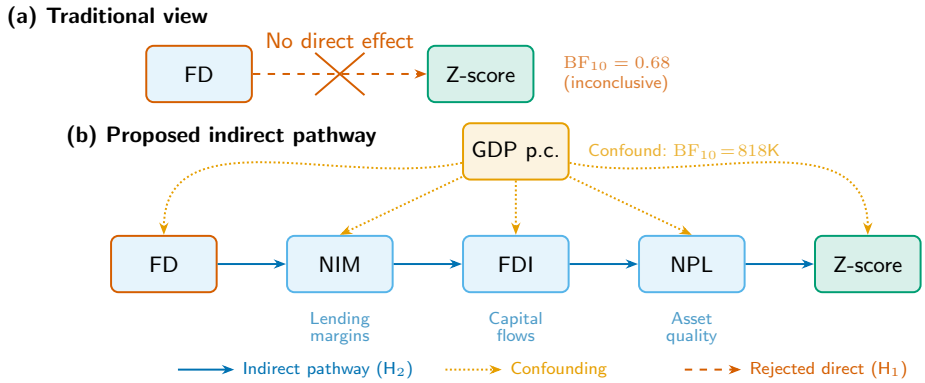

Fig. 1: Conceptual framework: hypothesized versus traditional model. (a) The traditional regression approach tests a direct effect of financial depth (FD) on bank stability (Z-score) and finds inconclusive evidence ($BF_{10} = 0.68$; supplementary median-split $BF_{10} = 0.16$). (b) Hypothesized indirect pathway FD $\rightarrow$ NIM $\rightarrow$ FDI $\rightarrow$ NPL $\rightarrow$ Z-score, with GDP per capita as a confounder ($BF_{10} = 818{,}156$). Note: this is the *a priori* hypothesis; empirical results show the indirect effect is not statistically significant.

model this as a direct, linear effect. If the true mechanism is indirect—operating through lending behavior, interest margins, or capital flows—then direct regression misses it. We argue that financial depth affects stability through an indirect pathway: FD $\rightarrow$ NIM $\rightarrow$ FDI $\rightarrow$ NPL $\rightarrow$ Z-score [29]. Fig. 1 contrasts the traditional view (Panel a, no direct effect) with the proposed indirect pathway (Panel b).

We test this using Bayesian network (BN) analysis on a panel of 10 ASEAN countries (2005–2020, $N = 285$ after MICE imputation). BNs learn conditional dependence structures from data [23,18]. We combine the PC algorithm [27], an expert-guided network with whitelist constraints [26], and bootstrap validation ($R = 1{,}000$).

Three findings emerge. First, Bayesian hypothesis testing yields inconclusive evidence for a direct FD–Z link ($BF_{10} = 0.68$), and the PC algorithm finds no direct edge—supporting H1. Second, GDP per capita is a dominant confounder ($BF_{10} = 818{,}156$); after conditioning, financial depth shows a negative association with stability ($\hat{\beta} = -0.130$, $t = -6.79$). Third, the indirect pathway exists structurally but transmits a negligible effect ($\hat{\theta} = 6.12 \times 10^{-4}$, 95% CI contains zero); H2 is not supported. This paper contributes by identifying GDP confounding as a primary source of inconsistency in this literature and demonstrating that structural pathway stability does not guarantee quantitative significance [1].

2 Literature Review

2.1 Financial Depth and Banking Stability

Finance begins to hurt growth when private credit exceeds 80–100% of GDP [3,20]. Moral hazard links this to banking: deposit insurance weakens market discipline, encouraging riskier lending [12,2,1]. We use private credit to GDP (GFDD.DI.12) as our financial depth (FD) measure. The Z-score combines profitability, capitalization, and return volatility [19]:

$$Z = \frac{\mathrm{ROA} + \mathrm{E/A}}{\sigma(\mathrm{ROA})} \tag{1}$$

NPL isolates credit risk [8], NIM indicates lending margins [5], and FDI captures competitive pressure from foreign capital [10].

2.2 Bayesian Networks and Mediation

Mediation analysis decomposes total effects into direct and indirect components [29]. BNs integrate mediation with structure discovery via DAGs [23,18]. The PC algorithm [27] removes edges between conditionally independent variables; Hill-Climbing (HC) maximizes a BIC-g score with expert constraints [26]. For our four-step pathway, the indirect effect is:

$$\mathrm{IE} = \beta_{\mathrm{FD}\to\mathrm{NIM}} \times \beta_{\mathrm{NIM}\to\mathrm{FDI}} \times \beta_{\mathrm{FDI}\to\mathrm{NPL}} \times \beta_{\mathrm{NPL}\to Z} \tag{2}$$

2.3 Research Gap and Hypotheses

No study has applied BNs to test whether financial depth transmits to banking stability through lending channels [3,22]. GDP confounding remains understudied [9]. We propose:

H1. The PC algorithm at $\alpha = 0.05$ does not identify a direct edge FD $\to Z$.

H2. The total indirect effect through FD $\to$ NIM $\to$ FDI $\to$ NPL $\to Z$ is statistically distinguishable from zero (bootstrap CI excludes zero).

3 Methodology

3.1 Data and Variables

We compile a panel dataset covering 10 ASEAN countries (2005–2020). Multiple imputation (MICE, $m = 5$) yields $N = 285$ country-year observations across 8 variables [6]. Variables are defined in Sect. 2.1 (GDP per capita, government debt, and CPI complete the eight-variable set); data come from World Bank GFDD [31], WDI [32], and IMF FSI [17]. We adopt Bayes factors (BF_{10}) with the JZS prior ($r = 0.707$) [30,25].

3.2 Bayesian Network Framework

A Bayesian network $\mathcal{B} = (\mathcal{G}, \Theta)$ is a DAG where nodes represent random variables and edges encode direct dependencies; the joint distribution factorizes as $P(X_1, \ldots, X_p) = \prod_i P(X_i \mid \mathrm{Pa}(X_i))$ [23]. The d-separation criterion tests conditional independence, allowing us to assess whether FD and Z are independent given mediators [18]. In our setting, $p = 8$ variables (Sect. 3.1). We use BN structure learning rather than fixed-effects regression because the objective is *structure discovery* [26]. To guard against pooling bias, we supplement with fixed-effects de-meaned BN, cluster bootstrap, and cluster-robust standard errors (Supplementary Material).

3.3 Structure Learning

We use the order-independent PC algorithm [27,11] as the primary data-driven method. Phase 1 recovers the skeleton by testing conditional independence for each pair (X_i, X_j) given conditioning sets of increasing size, removing an edge when the partial correlation test yields $p > \alpha = 0.05$ ($N = 285$). Phase 2 orients edges by identifying v-structures and applying Meek's rules.

3.4 Expert-Guided Bayesian Network

We complement the PC algorithm with an expert-guided Hill-Climbing (HC) algorithm that maximizes the BIC-g score subject to whitelist and blacklist constraints encoding domain knowledge [16]. The whitelist forces the hypothesized indirect pathway (FD→NIM→FDI→NPL→Z) and GDP's exogenous status; the blacklist prevents reverse causation and implausible feedback. The whitelist forces six edges—the hypothesized pathway plus GDP $\rightarrow Z$ and GDP $\rightarrow$ GovDebt—while the blacklist forbids reverse causation and implausible feedback. The remaining edges are unconstrained, allowing the HC algorithm to discover additional relationships from the data.

3.5 Path Coefficient Estimation

Given the learned DAG, we fit a Gaussian BN with linear conditional distributions [22], estimating each β_{ij} by OLS regression of X_i on its parents. The indirect effect through the mediation chain is:

$$\mathrm{IE}_{\mathrm{FD}\rightarrow Z} = \hat{\beta}_{\mathrm{FD}\rightarrow\mathrm{NIM}} \times \hat{\beta}_{\mathrm{NIM}\rightarrow\mathrm{FDI}} \times \hat{\beta}_{\mathrm{FDI}\rightarrow\mathrm{NPL}} \times \hat{\beta}_{\mathrm{NPL}\rightarrow Z} \tag{3}$$

Under H1, no direct FD–Z edge exists and the total effect equals the indirect effect [29].

3.6 Bootstrap Validation

Non-parametric bootstrap ($R = 1{,}000$) computes edge inclusion probabilities (EIP) and 95% percentile CIs [13,28]. Edges are classified as robust (EIP ≥ 0.85), moderate (0.50–0.85), or unstable (< 0.50). A cluster bootstrap, fixed-effects BN, and cluster-robust SEs address panel dependence (Supplementary Material).

3.7 Sensitivity Analysis

We conduct three robustness checks: (1) varying the PC algorithm's α over $\{0.01, 0.025, 0.05, 0.10, 0.15, 0.20\}$ to assess edge sensitivity; (2) computing partial correlations $\rho_{Z,\mathrm{FD}|\mathbf{M}}$ controlling for mediators, where ≈ 0 supports full mediation; and (3) leave-one-country-out (LOCO) analysis, re-estimating the BN for each of thirteen country-removal settings (ten ASEAN members plus three South Asian comparators) to guard against single-country influence.

3.8 Software Implementation

All analyses use R 4.5.2 with the `bnlearn` package v5.0 [26]. Replication scripts are in the Supplementary Material.

4 Data

Data come from the World Bank GFDD (depth, stability), WDI (GDP, FDI), IMF FSI (NPL, capital), and the WB/IADI Deposit Insurance Database, covering ten ASEAN member states (2005–2020). Sources were standardized, merged by country-year, and filtered to eight pathway variables. MICE ($m = 5$, predictive mean matching) with a 50% completeness filter yields $N = 285$ observations [6]. Little's MCAR test rejects completely random missingness ($p < 2.2 \times 10^{-16}$); we proceed under MAR. Variable definitions are in Sect. 3.1; descriptive statistics and cleaning details are in Supplementary Material. GDP per capita spans a 94-fold range across ASEAN, providing identifying variation. FDI measures aggregate inflows, not banking-specific flows; measurement error may attenuate path coefficients (see Limitations).

5 Results

The descriptive statistics are in Supplementary Material Table S2. The sample comprises $N = 285$ country-year observations (MICE, $m = 5$) from ten ASEAN states (2005–2020).

5.1 Bayesian Hypothesis Testing

We use Bayes factors (BF_{10}) following [30]: $BF_{10} < 1/3$ = evidence for H_0; 1/3–3 = inconclusive; > 10 = strong for H_1. Table 1 reports correlations for the key variable pairs.

Three patterns stand out: (1) Z–NPL ($r = -0.36$, $BF_{10} = 10.33$): strong evidence that NPL is inversely linked to stability [4]; (2) GDP–Z ($r = 0.63$, $BF_{10} = 818{,}156$): extreme evidence for GDP as a dominant confounder; (3) FD–Z ($r = -0.23$, $BF_{10} = 0.68$): inconclusive, resolving only after GDP adjustment.

A supplementary median-split t-test ($BF_{10} = 0.16$) provides moderate evidence for the null, bracketing the direct FD–Z evidence between inconclusive and null [21]. Data distributions and the correlation structure are visualized in Supplementary Material Figs. S1–S4.

Table 1: Bayesian Correlation Analysis

Pair	**Not.**	r	p	BF_{10}	**Evidence**
$Z \leftrightarrow$ NPL	$\rho_{Z,N}$	-0.36	0.004	10.33	Strong H_1
$Z \leftrightarrow$ FD	$\rho_{Z,F}$	-0.23	0.07	0.68	Inconclusive
GDP $\leftrightarrow Z$	$\rho_{G,Z}$	$+0.63$	4×10^{-8}	818,156	Extreme H_1
FD $\leftrightarrow$ GovD	$\rho_{F,G}$	$+0.08$	0.52	0.16	Mod. H_0

5.2 Conditional Independence Structure

The PC algorithm ($\alpha = 0.05$) discovers twelve edges (Table 2; full list in Supplementary Material).

Table 2: Selected PC Algorithm Edges ($\alpha = 0.05$, 8 vars). Eight of twelve pathway-relevant edges shown.

#	**Node 1**	**Node 2**	**Interpr.**
1	Z	GDP	Macro $\rightarrow$ stab.
2	Z	GovDebt	Fiscal $\rightarrow$ solv.
3	NPL	FDI	Risk via FDI
4	**FD**	**NIM**	**Depth$\rightarrow$margins**
5	FD	CPI	Depth$\sim$inflation
6	NIM	FDI	Margins$\rightarrow$flows
7	GDP	GovDebt	Income$\rightarrow$fiscal
8	GDP	FDI	Dev.$\rightarrow$FDI

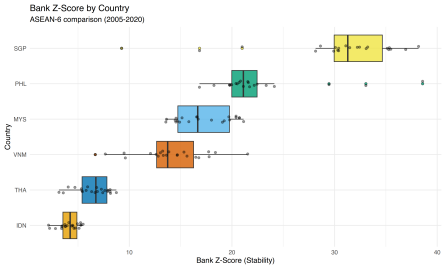

Fig. 2: Bank Z-score distribution by ASEAN country (2005–2020). The 94-fold GDP range (IDN≈5 vs. SGP≈32) illustrates why GDP confounding dominates.

H1 is supported: no direct FD–Z edge exists, consistent with the inconclusive-to-null Bayesian evidence. The algorithm does identify FD–NIM, NIM–FDI, and NPL–FDI, forming the skeleton of the hypothesized indirect pathway.

The D-separation analysis confirms that Z and FD are **not** d-separated: the indirect pathway through NIM, FDI and NPL remains open. We estimate the expert-guided BN using the whitelist and blacklist constraints (Sect. 3.4). The resulting DAG (Supplementary Material Fig. S1) retains all whitelisted edges plus data-driven edges (BIC $= -494.7$ vs. -490.9 unconstrained).

5.3 Path Coefficient Estimation

We estimate path coefficients along the hypothesized indirect pathway $FD \rightarrow NIM \rightarrow FDI \rightarrow NPL \rightarrow Z$-score using the fitted Gaussian Bayesian network. Each coefficient represents the expected change in the child variable per unit change in the parent.

GDP-Adjusted Association Estimates Table 3 reports GDP-adjusted partial association estimates from OLS regression adjustment, following [24]. These

estimates are observational: without random assignment or an instrumental variable, they reflect conditional associations rather than identified causal effects [14]. Unmeasured confounders—including institutional quality, regulatory stringency, and political risk—may bias these estimates.

Table 3: GDP-Adjusted Regression Coefficients ($N = 285$, OLS)

Regressor	Ctrl	$\hat{\beta}$	t	
$FD \to Z$	GDP	-0.130	-6.79	***
$FD \to Z$	GDP+GD	-0.037	-2.17	*
$FD \to NPL$	GDP	-0.010	-1.43	
$NIM \to NPL$	GDP	-0.200	-0.61	

$^{***}p < 0.01$; $^{*}p < 0.05$.

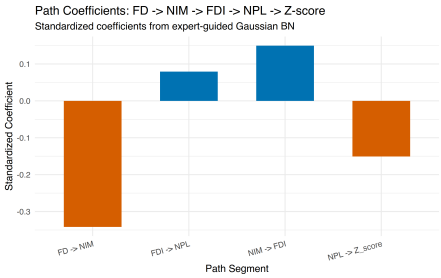

Fig. 3: Standardized path coefficients along the indirect pathway $FD \to NIM \to FDI \to NPL \to Z$. The first and last segments are negative; middle segments are positive but small.

After GDP control, FD has a significant negative association with Z ($\hat{\beta} = -0.130$, $t = -6.79$); adding GovDebt attenuates to -0.037 ($t = -2.17$, marginal). The CI test rejects $FD \perp\!\!\!\perp Z \mid$ mediators ($p = 1.5 \times 10^{-5}$), suggesting partial mediation at best. A supplementary non-linearity test reveals an inverted-U consistent with [3].

Path Coefficients Along the Indirect Pathway The standardized path coefficients along the indirect pathway are individually modest: $\hat{\beta}_{FD \to NIM} = -0.342$, $\hat{\beta}_{NIM \to FDI} = +0.150$, $\hat{\beta}_{FDI \to NPL} = +0.079$, $\hat{\beta}_{NPL \to Z} = -0.151$ (Supplementary Material Table S3). The four-step multiplication compounds attenuation. The total indirect effect is estimated as the product of path coefficients:

$$\begin{aligned}\hat{\theta}_{\text{indirect}} &= \hat{\beta}_{FD \to NIM} \times \hat{\beta}_{NIM \to FDI} \times \hat{\beta}_{FDI \to NPL} \times \hat{\beta}_{NPL \to Z} \\ &= (-0.342)(0.150)(0.079)(-0.151) = 6.12 \times 10^{-4} \quad (4)\end{aligned}$$

The four-step IE is negligibly small (≈ 0.0006); the attenuation cascade from $|-0.342|$ to $|+0.0006|$ is shown in Fig. 3.

5.4 Bootstrap Validation

Table 4 reports bootstrap EIPs ($R = 1{,}000$) for the indirect pathway.

EIP = 1.0 for all four pathway edges is mechanical: the whitelist forces inclusion. The unconstrained $NIM \to FDI$ strength is only 0.040 (Supplementary Material). The pathway's structural stability is an artifact of the whitelist, not an empirical discovery [26].

The bootstrap 95% CI for the indirect effect is $[-0.00020, 0.00219]$, containing zero. **H2 is not supported**: the point estimate (6.12×10^{-4}) is economically negligible and not statistically significant (Fig. 4).

Table 4: Bootstrap EIP ($R = 1{,}000$). All edges whitelisted; EIP = 1.0 is mechanical.

Edge	Str.	Dir.
$FD \rightarrow NIM$	1.00	1.00
$NIM \rightarrow FDI$	1.00	1.00
$FDI \rightarrow NPL$	1.00	1.00
$NPL \rightarrow Z$	1.00	1.00

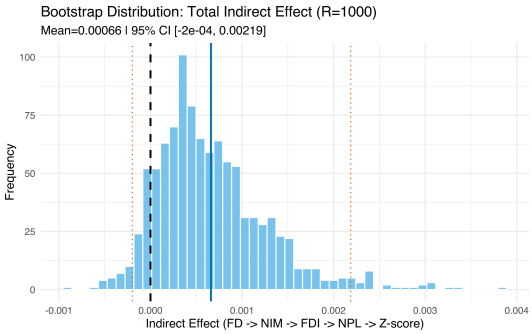

Fig. 4: Bootstrap distribution of the total indirect effect ($R = 1{,}000$). The 95% CI $[-0.0002, 0.0022]$ contains zero (dashed line); H2 is not supported.

5.5 Sensitivity Analysis

Conditional independence tests confirm FD and Z are not d-separated given the mediators ($p = 1.5 \times 10^{-5}$), suggesting partial mediation at best. GDP acts as a suppressor: the FD–Z partial correlation strengthens from -0.008 to -0.163 after GDP control. Leave-one-country-out (LOCO) analysis across thirteen country-removal settings (ten ASEAN plus three South Asian comparators) shows the indirect effect is negative in 12/13 settings (range: -0.00453 to -0.00011), contrasting with the small positive full-sample estimate ($+6.12 \times 10^{-4}$) and indicating sign instability near zero rather than directional robustness. A cluster bootstrap ($R = 1{,}000$, resampling countries) renders FD non-significant ($p = 0.56$), while GDP ($p = 0.003$) and GovDebt ($p < 0.001$) remain robust predictors. Detailed sensitivity analyses, partial correlation tables, LOCO results, and fixed-effects de-meaned BN results are provided in the Supplementary Material.

6 Discussion

Fig. 5 summarizes how four analytical methods converge on two conclusions: GDP confounds the FD–Z relationship, and the indirect pathway transmits a negligible effect.

6.1 GDP Confounding Masks the True FD Effect

The raw FD–Z correlation ($r = -0.23$, $BF_{10} = 0.68$) is inconclusive, but GDP-adjusted regression reveals $\hat{\beta} = -0.130$ ($t = -6.79$). Wealthier ASEAN economies have both deeper financial markets and more stable banks, creating confounding bias [23]. The extreme GDP–Z Bayes factor ($BF_{10} = 818{,}156$) and the PC algorithm's placement of GDP at the center of the conditioning set confirm this. Adding GovDebt attenuates the coefficient to -0.037 ($t = -2.17$), suggesting partial transmission through the sovereign-bank nexus [1].

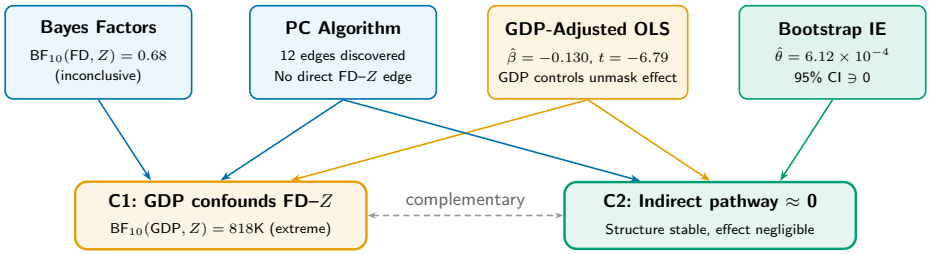

Fig. 5: Evidence Triangulation Summary. Four analytical methods—Bayes factors, the PC algorithm, GDP-adjusted regression, and bootstrap indirect effect estimation—converge on two complementary conclusions: (C1) GDP per capita confounds the FD–Z-score relationship, masking a significant negative association; (C2) the hypothesized four-step indirect pathway is structurally present but transmits a negligible effect.

However, cluster-robust inference renders FD non-significant ($p = 0.56$), so practical significance depends on how within-country dependence is treated.

6.2 The Indirect Financial Depth Mechanism

H2 is not supported under current measurement. The total indirect effect through FD $\rightarrow$ NIM $\rightarrow$ FDI $\rightarrow$ NPL $\rightarrow$ Z-score is negligible ($\hat{\theta} = 6.12 \times 10^{-4}$, 95% CI $[-0.0002, 0.0022]$). The four-step product of individually modest coefficients compounds to near zero, while the GDP-adjusted regression (-0.130, $t = -6.79$) confirms a real negative association operating through channels other than the pre-specified mediation chain. Shorter pathways (e.g., FD $\rightarrow$ NPL $\rightarrow$ Z) transmit effects more than 100 times larger [2].

6.3 Comparison with Existing Literature

Our finding of no direct FD–stability link does not contradict [12]—the mechanism operates through indirect channels. The GDP confounding result quantifies concerns by [2], while the pathway connects to [4] on NPL–stability links, though perfect EIPs result from whitelist constraints (Supplementary Material). The Bayesian framework quantifies evidence *for* H_0 rather than merely failing to reject it [30].

6.4 Policy Implications

These findings suggest hypotheses—not prescriptions—for ASEAN banking regulation [7]. The extreme GDP–Z association implies that supervisory frameworks tracking only aggregate credit-to-GDP ratios may miss early-warning signals, and the 94-fold GDP range suggests context-specific calibration [1]. Confirmation requires direct deposit insurance data and quasi-experimental designs.

6.5 Limitations

Nine limitations qualify these results: (1) sample size limits Bayesian power; (2) missing data assumes MAR; (3) causal identification requires untestable assumptions; (4) panel heterogeneity not modeled with FE/RE; (5) reverse causality cannot be ruled out; (6) contemporaneous BN cannot capture lagged transmission; (7) FDI proxy measures aggregate flows; (8) Bayes factor prior sensitivity; (9) financial depth proxy does not directly measure deposit insurance design. Detailed discussion of each limitation is provided in the Supplementary Material.

7 Conclusion

This study investigated whether financial depth (private credit to GDP) is associated with banking instability in ASEAN, and whether the association operates through the indirect lending pathway FD $\rightarrow$ NIM $\rightarrow$ FDI $\rightarrow$ NPL $\rightarrow$ Z-score, using Bayesian network analysis and hypothesis testing on $N = 285$ country-year observations (MICE, $m = 5$) from ten ASEAN states (2005–2020).

Two principal findings emerge. First, GDP per capita confounds the FD–stability relationship ($BF_{10} = 818{,}156$). The unconditional association is weak ($BF_{10} = 0.68$), but GDP-adjusted regression reveals $\hat{\beta} = -0.130$ ($t = -6.79$) under i.i.d. standard errors, though cluster-robust inference attenuates this to non-significance. Second, the four-step indirect pathway is structurally present (retained in all $R = 1{,}000$ bootstrap replications, noting whitelisted edges are forced by design) but economically negligible ($\hat{\theta} = 6.12 \times 10^{-4}$, 95% CI contains zero); **H2 is not supported under current measurement**. This carries a methodological lesson: structural significance (edge stability) should not be conflated with quantitative significance (effect magnitude), especially when mediators are measured with aggregate proxies [15].

The study contributes along three dimensions: (1) GDP confounding quantified as a primary source of mixed results in the FD–stability literature [2]; (2) a combined BN and Bayesian hypothesis testing framework that discovers conditional dependence structures and provides principled evidence for H_0 [30]; and (3) evidence that multi-step mediation compounds attenuation, yielding near-zero transmitted effects even from individually meaningful coefficients. Future research should expand the sample, incorporate firm-level data with direct deposit insurance measures, and employ dynamic Bayesian networks to capture temporal feedback [24].

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During the preparation of this work, the authors used Claude (Anthropic) for data analysis pipeline development, statistical computation, and language editing. The authors reviewed and edited all AI-generated content and take full responsibility for the accuracy and integrity of the published work.

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